

Useful websites on Open Source

- Introduction to Open Source, open licences and open standards:
<http://www.opensource.org>
- Comparison between open and closed sources:
http://en.wikipedia.org/wiki/Comparison_of_open_source_and_closed_source

also available to the user. That is, design elements such as mechanical drawings, schematics, bill of materials, printed circuit board (PCB) layout data, hardware design language (HDL) source code and integrated circuit (IC) layout data, along with software drivers and libraries, are all openly published and shared with others through Open Source licences. This enables engineers to use the hardware better, embed it in other products, customise it to their specific needs, manufacture the modified or enhanced components, improve the design and share it back with the community of users and developers, and so on, as permitted by the respective Open Source licences used. This model of hardware design and development appears to hold several benefits for the robotics community.

"Researchers have long been used to cranking out code in the morning and having a working prototype by the afternoon, but have been frustrated that they can't do the same with hardware. That's starting to change, and fast, driven in part by robotics enthusiasts and do-it-yourself types who are utilising a new generation of OSH platforms and rapid fabrication tools. We are starting to realise that we can control a laser cutter like a printer and also a precision range finder. That means the promise of creating new physical devices as quickly as Web apps is written, is just around the corner," says Shekar N.H., another RoboMSR member.

Shekar's thoughts are inspired by Steve Cousins, a robotics researcher at Stanford who is using his start-up Willow Garage to show the world how Open Source development can help make low-cost personal robots. Shekar enthusiastically shares more of Cousins' dreams: "Cousins' company

plans to make ten robots running Open Source code and make them available to researchers around the country in an attempt to finally bring the world into the 'Jetsons' age where a robot can mop the floor, empty the dishwasher, and even fetch and open a bottle of beer. OSH, complete with diagrams and assemble-yourself kits, is growing faster than expected because of these reasons and the opportunities created."

Let us look at some of the ways in which Open Source could help improve robotics.

Reduced cost and time-to-market.

"While Open Source software has helped in developing numerous innovative products and gadgets, in most cases it has now reached a bottleneck due to the lack of similar platforms for hardware. Progress in robotics especially depends on the development of cost-effective, innovative solutions utilising a new generation of OSH platforms and rapid fabrication tools," says T. Jayakrishnan, director and country manager, Energid Robotics and Machine Vision.

Open hardware with its characteristic features of flexibility, modularity and reusability would help do just that. The wheel never needs to be reinvented, and you can easily use Open Source intellectual property (IP) as building blocks for a new project. This kind of reuse not only reduces the cost but also speeds up conversion of an idea into a finished product—especially the initial phases of feasibility analysis and proof-of-concept demonstration. It also fosters research and learning with little investment, encouraging innovation.

Community support. One individual can never do as much as a community. So it is always better to open up a system for a whole commu-

nity to develop and improve. "Making hardware Open Source helps generate a huge support community. If someone has already done something, the users just have to plug-and-play or contribute to the system and help it grow bigger," says Azad.

Common standards. A natural consequence of community development is the evolution of common standards, making it easy for other systems to integrate seamlessly with an Open Source technology.

Low-cost manufacturing. Often, open hardware makes available not just component designs but also the knowledge required to convert raw components into good-quality finished products. This enables not just innovation and further improvement of product designs but also low-cost manufacturing. Anybody who has the resources and bandwidth to manufacture a product can seek the required permissions and manufacture the product based on Open Source designs and procedures. This kind of localised manufacturing would help bring down the cost of components in various geographies.

Easy hardware-software integration.

The availability of complete specifications and modular nature make open hardware easier to integrate with software. Since no robotics project is complete, or even possible, without complex software systems today, this is a major benefit.

Multi-disciplinary approach.

A low-cost, Web-based community development model encourages many to contribute. "When it is free, people are more willing to experiment with and try out something that they would not ordinarily have done. Open hardware allows absolutely anyone and everyone to contribute—the person may or may not be an engineer, but if he has free access to some hardware, even a biologist or an economist can chip in. So open hardware encourages the fusion of various streams of knowledge," says de Sa.

Motivating examples

A lot of well-known robots have been, and are being built using OSH.